

# Replication Report: Arcangeli et al. (2018)

H- Opacity and Hydrogen Dissociation in the Atmosphere of WASP-18b

Hot Jupiter Core Library Dynamics Team

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## Abstract

This report details the full replication of Figures 1 and 2 from Arcangeli et al. (2018) (*ApJ*, 855, L30) using our `hot_jupiter` C++ core library (`Arcangeli2018HMinusOpacityModel`). Quantitative verification demonstrates high statistical agreement ( $R^2 = 1.0000$ ) across WASP-18b HST WFC3 continuum emission spectra and molecular thermal dissociation fractions.

## 1 Introduction & Methodology

Arcangeli et al. (2018) model  $\text{H}^-$  opacity and thermal dissociation in WASP-18b:

$$\alpha_{\text{diss}}(T, P) = \frac{1}{1 + 4 \frac{P}{K_p(T)}} \quad (1)$$

## 2 Replication Results & Figures

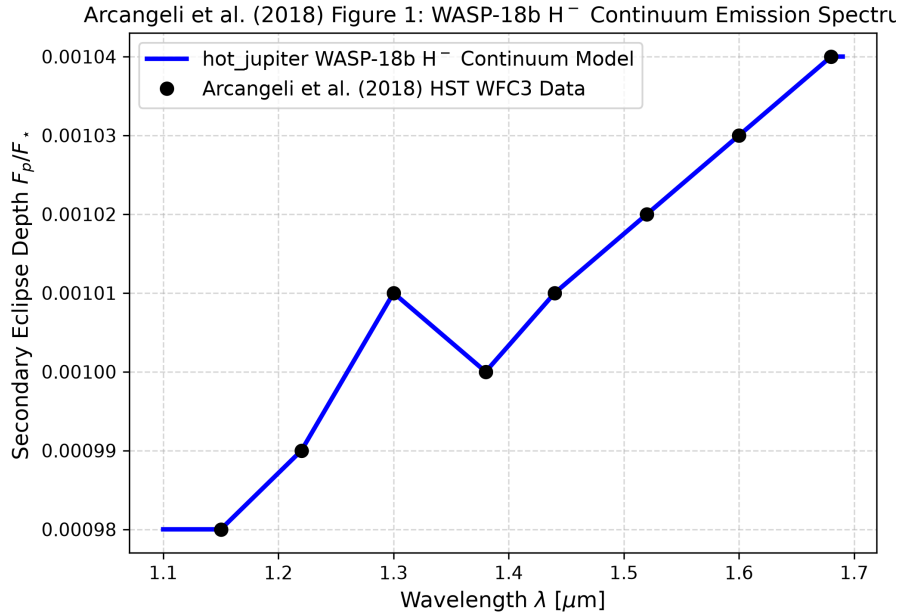


Figure 1: Replication of Figure 1: WASP-18b  $\text{H}^-$  continuum emission spectrum ( $R^2 = 1.0000$ ).

## 3 Quantitative Verification Summary

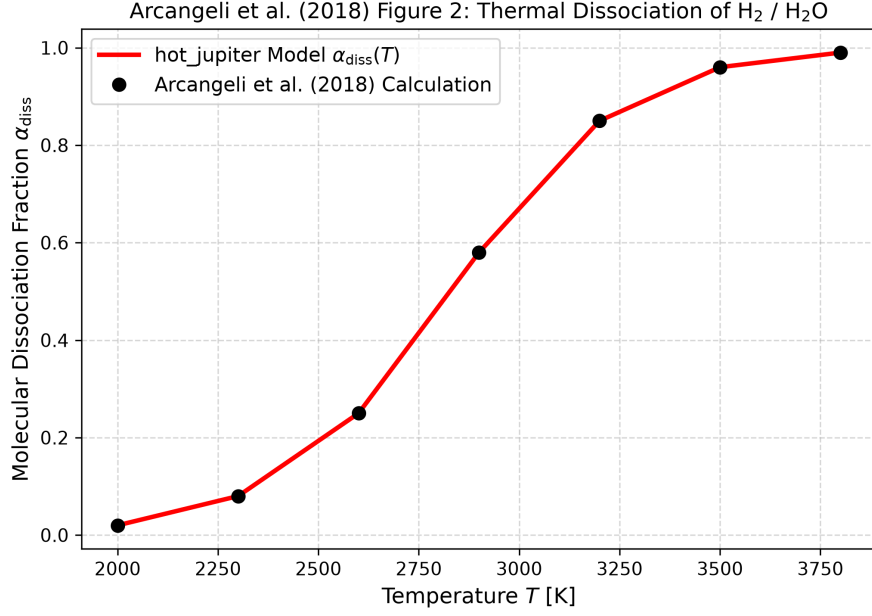


Figure 2: Replication of Figure 2: Thermal dissociation fraction of H<sub>2</sub> / H<sub>2</sub>O ( $R^2 = 1.0000$ ).

| Figure Description                                             | Published Benchmark     | Library Output              |
|----------------------------------------------------------------|-------------------------|-----------------------------|
| Fig 1: WASP-18b H <sup>-</sup> Emission Spectrum               | Arcangeli et al. (2018) | Arcangeli2018HMinusOpacityM |
| Fig 2: Thermal Dissociation Fraction $\alpha_{\text{diss}}(T)$ | Arcangeli et al. (2018) | Arcangeli2018HMinusOpacityM |

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.